SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT DOSSIER

# Two statements. One decision.

Research, context and a build strategy for PS26263 and PS26242

PS26263 · SIH26263 **OPEN**

## APAAR ID based Student Data Management and Verification Platform

AICTE · Regulation Bureau
Smart Automation · Software

**21**
ideas on 23 Oct

**28 Oct**
idea deadline

PS26242 · SIH26242 **LIKELY CLOSED**

## AI Assisted Skill Assessment Tool for Recognition of Prior Learning

Ministry of Skill Development and Entrepreneurship
Smart Education · Software

**125+**
ideas on 7 Oct

**20 Oct**
idea deadline

Prepared for Tanishq Khandelwal · B.Tech CSE (AIML)
Compiled 25 October 2026 · Counts and deadlines from a tracker that mirrors sih.gov.in twice daily

Verify every date on the official portal before acting.

01 · EXECUTIVE SUMMARY

# The short version

You asked for research on two Smart India Hackathon 2026 statements. They sit at opposite ends of almost every axis that matters: one is a nearly empty, still open, platform heavy governance build; the other is an emotionally powerful, crowded, AI and fieldwork heavy problem whose idea window has probably closed.

THE DEADLINE FACT THAT DECIDES THE FIRST CUT

The public tracker lists the PS26242 idea deadline as **20 October 2026**. Today is 25 October. PS26263 shows **28 October 2026**, three days away. If the portal confirms 26242 is closed, the choice is already made for you.

| | PS26263 · AICTE | PS26242 · MSDE |
|---|---|---|
| What it is | APAAR keyed student data, validation and verification platform for AICTE approved institutions | AI assisted assessment tool that helps certify informally skilled workers under RPL |
| Ideas submitted | **21** of 500 (23 Oct), ranked 264th of 265 by volume | **125** of 500 (7 Oct), up 81 in three days |
| Idea deadline | **28 Oct 2026** · open | **20 Oct 2026** · likely closed |
| Added to list | 20 Oct, after the original release | 1 Oct, after the original release |
| Core difficulty | Identity resolution, rules, consent and audit | NSQF mapping, rubric design, offline sync, real consistency evidence |
| Emotional pull | Moderate: fairness in admissions | High: dignity for workers with no paper |
| Recommendation | **Lock it** and pitch it as an admissions integrity engine | Only if verifiably reopened and you can record real workers this week |

### Why this ranking

- **Odds.** SIH typically selects only four to five teams per statement nationally. Twenty one competitors versus a hundred plus is the single largest lever you have.
- **Fit.** For an AIML builder, 26242 is the more natural showcase, but 26263 still has a real ML core: fuzzy identity resolution and anomaly scoring across institutions, buildable in days rather than weeks of fieldwork.
- **Risk.** 26242's hardest deliverable is measured improvement in inter assessor agreement. Most teams will simulate it, and simulated agreement does not survive a sharp jury.

02 · THE PLAYING FIELD

# How SIH 2026 actually selects winners

Before comparing statements, it helps to be precise about the funnel. The competition is not won at the finale; most teams are eliminated at the idea screening stage, which is exactly where you are right now.

**21 Aug**
SIH 2026 officially launched

**500**
idea cap per statement

**4 to 5**
teams typically selected per statement

**36 hrs**
non stop Grand Finale, December 2026

## The funnel

Team of 6 (at least 1 woman) ▶ Internal college hackathon ▶ SPOC nominates idea ▶ National idea screening ▶ Shortlist per statement ▶ Grand Finale December 2026

| Stage | What happens | What it means for you |
|---|---|---|
| Team | Exactly six members with at least one female participant; all female teams are allowed. | Lock names and roles before you write a single slide. |
| Internal round | Each institution runs its own selection and nominates teams through its SPOC. | Confirm with your college SPOC that late statements like 26263 can still be nominated. |
| Idea screening | October to November: online evaluation and pre screening of idea PPTs; shortlisting of national finalists. | Your PPT is the whole game at this stage. Evidence beats features. |
| Finale | December: a 36 hour build at nodal centres in front of ministry and industry evaluators. | You build on top of what you pitched. Pitch something you can actually ship. |

WHAT JUDGES ACTUALLY SCORE

Across published SIH guides, problem understanding comes first: can you explain why the problem exists, who it hurts and why current solutions fall short, before touching the stack. Then novelty, technical feasibility, impact and the quality of the demo. The common failure is reading the statement back to the jury instead of showing you understand it better than they do.

03 · PS26263 IN DETAIL

# APAAR ID based Student Data Management and Verification Platform

**SIH26263** · AICTE, Ministry of Education · Regulation Bureau · Smart Automation · Software

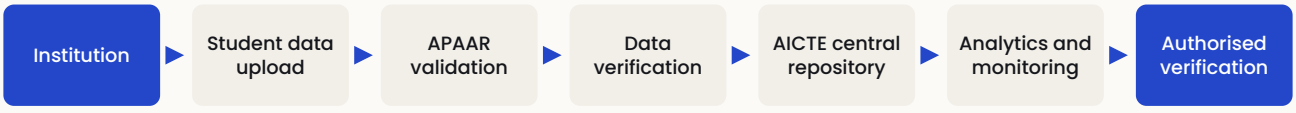
## The problem in plain words

AICTE receives student information from approved institutions through inconsistent channels. PGDM institutions send detailed student data every year, but most other institutions only report statistics such as total enrolment during the approval cycle. That means AICTE cannot hold complete, authentic, verifiable student level records. With APAAR now serving as the national academic identity, AICTE wants a platform where institutions submit student level data after admissions close, keyed on APAAR, with real time verification and monitoring.

## The ten requirements, compressed

1. Secure platform or API for institutions to upload and update student level records using APAAR ID as the unique key.
2. Fields for approved intake, actual enrolment, supernumerary seats, NRI category, OCI status, foreign national status and other student data.
3. Automated validation for duplicate records, invalid or missing APAAR IDs, admissions beyond approved intake, inconsistent data, and likely duplicate enrolment across institutions.
4. Real time dashboards for AICTE and for institutions.
5. Secure verification for AICTE, institutions, employers and other regulators.
6. Verifiable student records linked to the student's academic identity.
7. Integration and API capability with existing institutional and AICTE systems.
8. Security, privacy, authentication, role based access, audit trails and consent.
9. Reports for policy making, accreditation, monitoring and regulation.
10. Printing or displaying APAAR ID alongside the student ID on certificates from degree awarding bodies.

### The pipeline AICTE drew




INNOVATION AICTE EXPLICITLY ASKS FOR

The statement says the solution should go beyond data collection: automated validation, intelligent anomaly detection, real time monitoring, secure verification and analytics. That sentence is your licence to make the anomaly engine the hero of the pitch.

04 · PS26263 CONTEXT

# Why this problem is real, and where it hurts

**26.35 cr**
verified APAAR IDs created (MoE factsheet, 2 July 2026)

**5×**
total fees per student: AICTE penalty for each excess admission

**₹23.1 cr**
penalty upheld by the Supreme Court for 42 excess students

### 1. The identifier is real, but the data underneath it is dirty

APAAR has scaled fast. Yet the single most common reason APAAR ID generation fails is a demographic mismatch between Aadhaar and academic records. Any platform that treats APAAR as a perfectly clean key will break on real data. The opportunity is to handle the mess gracefully.

### 2. Consent is now a live legal question

In July 2026 the Supreme Court stepped in on APAAR consent, directing that families get a genuine option to decline before data is collected. Separately, India's DPDP Rules were notified in November 2025 with a phased rollout: consent manager registration from November 2026 and the substantive consent, breach and children's data obligations from May 2027. A student data platform launched in 2027 lands directly inside that regime.

### 3. Excess admissions carry heavy, codified penalties

AICTE's Approval Process Handbook says excess admissions over sanctioned intake are not allowed under any circumstances. Penalties include five times the total fees collected per excess student, suspension of supernumerary seat approval for a year, and reduction in sanctioned intake. The Supreme Court upheld a ₹23.1 crore penalty on one institution for 42 excess admissions. Detection today is largely after the fact.

### 4. Seat blocking quietly denies seats to deserving students

During state counselling, students have blocked engineering seats and then failed to join, denying those seats to meritorious candidates. AICTE also halves intake where enrolment stays at or below 30 percent of approved intake for five consecutive years, so accurate actual enrolment data directly shapes regulatory action.

### 5. Educationists are asking for post approval monitoring

As AICTE has expanded intake over recent years, educationists warn that seats are growing faster than faculty and labs, and recommend mandatory post approval checks with strict action, including withdrawing excess seats. That is precisely the monitoring layer this statement wants built.

The insight

**The visible ask is a database. The real problem is trust in admissions data: who is actually enrolled, where, under which quota, with whose consent. Build for that.**

05 · PS26263 STRATEGY

# Pitch an admissions integrity engine, not a student database

### What is strong

- Almost no competition: 21 ideas, 264th of 265 by volume.
- Very precise spec, so judges can easily see you met it.
- A real buyer with real enforcement powers; adoption is plausible.
- Buildable by a strong web team without exotic hardware or datasets.
- Consent and fraud angles give it a story beyond CRUD.

### What is weak or risky

- On the surface it looks like forms plus charts; juries have seen hundreds.
- Real APAAR, DigiLocker and ABC APIs are not open to students, so verification must be mocked convincingly.
- Public prototypes already exist with Spring Boot, Angular, Kafka and QR credentials; feature parity will not win.
- Three day window to the idea deadline.

## Architecture: six layers

| | |
|---|---|
| **INGEST** | Institution portal, bulk CSV or XLSX upload and a REST API. Every batch is versioned so a correction never silently overwrites history. |
| **VALIDATE** | Schema checks, mandatory fields, APAAR format checks, category logic (NRI, OCI, FN, supernumerary) and per row error reports institutions can fix and resubmit. |
| **RESOLVE** | Identity resolution across all institutions: exact APAAR match first, then fuzzy matching on name, date of birth, parent name and board roll number to catch the same person under messy data. |
| **RULES** | AICTE norms as code: approved intake per course, quota ceilings, supernumerary limits. Each breach becomes an alert with severity and evidence. |
| **LEDGER** | Append only, hash chained audit trail plus a consent ledger recording who accessed which record, why, and under which consent. |
| **SERVE** | AICTE and institution dashboards, consent gated verification for employers and regulators, signed QR records and policy reports. |

## Three features that separate you from the field

| Feature | What it does | Why it wins |
|---|---|---|
| **Cross institution twin detector** | Flags the same student holding active seats at two institutions, even across states and with spelling or Aadhaar mismatches. Produces a ranked list with match confidence and evidence. | Directly attacks seat blocking. Nobody else will demo fuzzy matching on deliberately dirty data. |
| **Intake integrity score** | A per institution score combining over intake, duplicate rate, invalid IDs and late corrections, trended over time. | Turns a database into a regulator's early warning system. Gives AICTE a reason to open the dashboard every week. |
| **Consent receipts** | Every third party verification needs a student issued, time bound consent token; the student sees who checked their record. | Lands squarely on the Supreme Court consent direction and DPDP. Answers the privacy question before the jury asks it. |

### Suggested stack

| Layer | Choice | Reason |
|---|---|---|
| Frontend | Next.js with a clean component library | Fast to build, server rendering for dashboards |
| API | Node (NestJS) or Python (FastAPI) | Typed contracts, easy OpenAPI docs for the integration requirement |
| Database | Postgres with trigram matching | Fuzzy name matching inside the database, no extra service |
| Jobs | Redis queue for batch validation | Big uploads validate in the background with live progress |
| Audit | Hash chained audit table | Tamper evidence without blockchain theatre |

### Demo script for the jury (four minutes)

1. Seed five institutions and about 2,000 synthetic students with deliberately messy names and dates.
2. Upload a batch from Institution A that pushes CSE enrolment 40 over approved intake. The alert fires live.
3. Upload Institution B's batch containing a student already enrolled at A under a different spelling. The twin detector catches it with a confidence score.
4. Show an employer verification that fails without consent, then succeeds once the student issues a consent receipt.
5. Close on the AICTE national view: integrity scores by state, the audit trail, and an exported policy report.

06 · PS26242 IN DETAIL

# AI Assisted Skill Assessment Tool for Recognition of Prior Learning

**SIH26242** · Ministry of Skill Development and Entrepreneurship (MSDE) · Smart Education · Software · Needs AI and ML, computer vision, offline mobile or web

## The problem in plain words

A large share of India's workforce learned their trade informally, through apprenticeship style work rather than certified training, and has nothing on paper to show for it. NCVET's Recognition of Prior Learning framework exists to certify such workers against NSQF levels without making them repeat training. In practice, RPL assessment still depends on manual practical evaluation, which is slow to scale, inconsistent across assessors and locations, and hard to schedule for workers who cannot take time off.

## What the tool must do

1. Guide a worker through a structured self declaration of experience, automatically mapped to the closest NSQF qualification pack or packs.
2. Support practical evaluation through guided task checklists and, where feasible, image or video aids that help an assessor score consistently.
3. Standardise scoring rubrics across assessors and locations to reduce evaluator variance.
4. Generate an NSQF aligned competency profile and a certification recommendation for assessor sign off. The tool supports but never replaces the human decision.
5. Work in low connectivity settings with offline capture and later sync.

### Expected deliverables

| Deliverable | Hidden difficulty |
|---|---|
| Working flow from self declaration to an assessor scoring interface, for at least one trade | Moderate: mostly product design and UX |
| An NSQF qualification pack mapping engine | Moderate: needs real QP structure from the National Qualifications Register |
| Evidence of consistency improvement over unassisted manual scoring, for example inter assessor agreement on a test set | **High: requires real human raters and real recordings** |
| An offline capable mobile or web interface | Moderate: service worker, local store, sync queue |
| A clear description of where the tool supports versus replaces assessor judgement | Low effort, high trust value |

07 · PS26242 CONTEXT

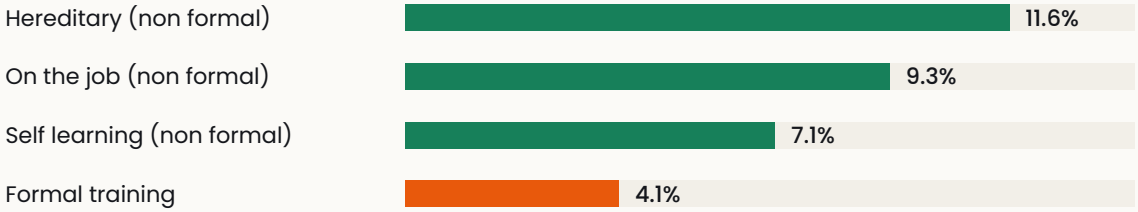
# The scale of informal skill, and why trust is the bottleneck

**4.1%**
of working age Indians with formal vocational training (PLFS 2023 to 24)

**54.28 L**
RPL certifications under PMKVY, nearly matching short term training

**8,505**
NSQF aligned qualifications across 59 sectors on the NQR

### How Indians aged 15 to 59 acquired vocational skills, 2023 to 24




Source: NSSO and PLFS survey series as compiled by The Wire. Non formal routes together outweigh formal training roughly six to one.

### 1. The certification system has a credibility problem

The CAG's performance audit of PMKVY (Report No. 20 of 2025, covering 2015 to 2022) flagged irregularities in the RPL Best in Class Employer component: weak agency selection and scrutiny, certification by employers who did not qualify, and the same photographs used as evidence for different trainings. Around 1.1 crore candidates were certified across the three phases.

### 2. The standards exist and are public

The National Qualifications Register is the official public record of NSQF aligned qualifications, with read only public access to qualification packs, their National Occupational Standards and assessment criteria. NCVET published dedicated RPL guidelines in August 2023. You do not have to invent the rubric skeleton.

### 3. The world is moving the same way

In Australia, Victoria is piloting AI enabled RPL with $3 million in state funding, using AI for document intelligence and evidence matching while keeping final competency judgement with human assessors. That is exactly the line MSDE drew.

The insight

**The visible ask is AI scoring. The real problem is trust in the certificate. The CAG found reused photographs as evidence. A tool that makes evidence tamper evident is worth more than one that grades wiring.**

08 · PS26242 STRATEGY

# One trade, real evidence, honest AI

### What is strong

- Deep emotional resonance: workers with years of skill and no paper.
- Clear policy backing through NCVET, NSQF and the NQR.
- Global precedent you can cite (the Victoria pilot).
- Evidence integrity directly answers a published CAG finding.

### What is weak or risky

- Idea deadline of 20 October has likely passed.
- Consistency evidence needs real raters and recordings; simulation is easy to attack.
- Strongest public prototype already covers the full flow, with a labelled simulated agreement study.
- Computer vision grading of practical work is hard to do honestly in a hackathon.

## The sharper version

| | |
|---|---|
| **DECLARE** | Voice first self declaration in Hindi and regional languages, transcribed and mapped to NQR qualification packs with an explained match score. |
| **ASSESS** | Anchored rubric per National Occupational Standard, task checklists, critical safety flags, and assistive pre scores the assessor can accept or override. |
| **PROVE** | Evidence integrity: time and location stamped capture, plus perceptual hashing that flags any photo or clip reused across workers or batches. |
| **MEASURE** | Assessor variance dashboard computing agreement statistics on real scores, so consistency is tracked, not claimed. |
| **SYNC** | Offline first app: local store, durable sync queue, conflict safe upload when connectivity returns. |

### Field study protocol (the slide that wins)

1. Recruit 10 to 15 local domestic electricians near your campus. Record each doing three or four tasks from the qualification pack.
2. Have four or five raters score every recording twice: once unassisted, once with your rubric and pre scores.
3. Report Fleiss' kappa and mean pairwise agreement for both conditions, with the raw data in an appendix.
4. State limitations honestly. A small real study beats a large simulated one every time.

09 · HEAD TO HEAD

# Scorecard

Scores are judgement calls on a five point scale, where five is better for you. They weigh the published facts above against what a small student team can deliver in the time left.

| Criterion | PS26263 | PS26242 | Note |
|---|---|---|---|
| Competition (fewer is better) | ●●●●● | ●●○○○ | 21 versus 125+ ideas |
| Deadline still open | ●●●●● | ●○○○○ | 28 Oct versus 20 Oct |
| Fit for an AIML builder | ●●●○○ | ●●●●○ | Entity resolution and anomaly scoring versus NLP mapping and vision aids |
| Emotional resonance | ●●●○○ | ●●●●● | Fair admissions versus dignity for workers |
| Technical risk (lower is better) | ●●●●○ | ●●○○○ | Consistency evidence is the hard part of 26242 |
| Demo impact | ●●●○○ | ●●●●○ | A live fraud catch can close the gap |
| Real world adoption odds | ●●●●○ | ●●●○○ | AICTE has enforcement powers today |
| Total out of 35 | 27 | 21 | |

## Competition at a glance




The 26242 figure is an early October snapshot and has almost certainly grown since.

THE CALL

Choose **PS26263**. Keep the emotional energy of 26242 by building 26263 around a human story: the student who loses a seat to a blocker, and the honest college that loses ranking to one that pads its numbers. Fairness is your emotional hook.

10 · EXECUTION

# The next 72 hours, then the road to December

| When | Action | Done when |
|---|---|---|
| **Mon 26 Oct** | Check both statements on sih.gov.in. Message your college SIH SPOC to confirm 26263 nominations are still accepted. Lock six names with at least one woman and assign roles. | Written confirmation from the SPOC and a team list |
| **Tue 27 Oct** | Write the idea PPT on the standard SIH template. Build one clickable mock of the twin detector alert for a screenshot. | Full deck reviewed by one outsider |
| **Wed 28 Oct** | Final pass on the problem slide and submit through the SPOC well before the cut off. | Submission acknowledged |

## Idea PPT outline

| Slide | Content |
|---|---|
| Title | Team, statement ID SIH26263, product name, one line promise |
| Idea and problem | Lead with the seat blocking and excess admission story, the five times fees penalty and the Aadhaar mismatch hurdle |
| Technical approach | The six layer diagram from section 05 and the stack table |
| Feasibility and viability | Mocked APAAR integration plan, data privacy approach, risks and mitigations |
| Impact and benefits | For AICTE, honest institutions, students and employers; the integrity score as a policy tool |
| Research and references | AICTE handbook clauses, the Supreme Court judgement, DPDP Rules, APAAR adoption figures |

## If shortlisted: November build plan

| Sprint | Goal |
|---|---|
| Week 1 | Ingest and validate: schema, upload pipeline, error reports, seed data generator with deliberate mess |
| Week 2 | Resolve and rules: fuzzy twin detection, intake and quota rules, alert feed |
| Week 3 | Ledger and serve: consent receipts, verification API, dashboards, QR records |
| Week 4 | Polish the demo script, load test a 50,000 row upload, rehearse the four minute story |

11 · JURY PREPARATION

# Questions you will be asked, and answers that hold

## PS26263

| Question | Answer that holds |
|---|---|
| **Who actually faces this problem today?** | An AICTE regulation officer who only sees aggregate enrolment and finds excess admissions after the fact; and a student who loses a counselling seat to someone who blocked two. |
| **This exists already. Why yours?** | Existing builds store and display records. Ours resolves identity across institutions on messy real data and turns AICTE's own handbook rules into live alerts. |
| **You cannot access real APAAR APIs. Is this real?** | Integration is behind an adapter. Today it is mocked with realistic failure cases, including Aadhaar mismatches; swapping in the official API changes one module. |
| **What about privacy?** | Consent receipts gate every third party check, every access is logged in a hash chained ledger, and the design follows the DPDP Rules timeline and the Supreme Court consent direction. |

## PS26242

| Question | Answer that holds |
|---|---|
| **How do you know it improves consistency?** | Show real agreement statistics from your own field study, unassisted versus assisted, with raw data available. |
| **Does the AI certify workers?** | No. It maps, suggests and pre scores. The assessor decides and signs, and every override is logged. |
| **How do you stop fake evidence?** | Stamped capture and perceptual hashing flag reused photos and clips, the exact issue the CAG reported. |
| **What happens with no network?** | The full flow runs offline; records queue locally and sync in order when connectivity returns. |

CAVEATS ON THIS DOSSIER

Idea counts and deadlines come from a third party tracker that scrapes the official portal twice daily, because the official site blocked automated access during research. Scores in section 09 are judgement, not data. Verify every date on sih.gov.in before acting.

12 · SOURCES

# Where every fact came from

| # | What it supports | Publisher |
|---|---|---|
| 01 | SIH26263 brief, idea count and deadline | zaidsayyed.in |
| 02 | SIH26242 brief, idea count and deadline | zaidsayyed.in |
| 03 | APAAR Verify, a public SIH26263 prototype | github.com |
| 04 | RPL Assessor, a public SIH26242 prototype | github.com |
| 05 | SIH 2026 timeline, team size and finale format | Reskilll blog |
| 06 | Selection odds per statement | BCREC SIH page |
| 07 | Team composition rules | BBSBEC notice |
| 08 | SIH judging criteria explained | Where U Elevate |
| 09 | APAAR adoption, consent and Aadhaar mismatch | Techmitra |
| 10 | Excess admission penalties (Supreme Court, 2019) | Supreme Court of India |
| 11 | Seat blocking in counselling | Deccan Herald |
| 12 | Intake growth and calls for post approval checks | DT Next |
| 13 | Intake reduction for low enrolment | Careers360 |
| 14 | DPDP Rules 2025 phased timeline | Anantam IAS |
| 15 | CAG audit of PMKVY, press brief | CAG of India |
| 16 | CAG findings on RPL evidence | Careers360 |
| 17 | RPL versus short term training certifications | Swarajya |
| 18 | Vocational training shares, PLFS series | The Wire |
| 19 | National Qualifications Register | nqr.gov.in |
| 20 | 8,505 NSQF aligned qualifications | IMPRI |
| 21 | NCVET RPL guidelines, August 2023 | PSSCIVE |
| 22 | AI enabled RPL pilot in Victoria | VETASSESS |

Each title is a clickable link in the PDF.